Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: **CACS252** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. Compute the root of equation $x^2 - 5x + 6 = 0$ in the vicinity of $x = 5$ using Newton-Raphson method.

3. Estimate the value of $\ln(2.5)$ using Newton-Gregory forward difference formula given the following data:X1.02.03.04.0Ln(x)0.00.69311.09861.3863

4. Write an algorithm to compute integration using Simpson's 1/3 rule. Evaluate integration of $\int_0^3 (3x^2 + 2x - 5) \, dx$ using Simpson's 1/3 rule.

5. What is meant by ill-conditioned system? Find the Cholesky decomposition of the following matrix: $$\begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

6. Use the classical RK method to estimate $y(0.4)$ of the equation $\frac{dy}{dx} = x^2 + y^2$ with $y(0) = 0$, assume $h = 0.2$.

7. Solve the Poisson equation $\nabla^2 f = 2x^2 y^2$ over the square domain $0 \leq x \leq 3 \text{ and } \quad 0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.

8. Write a short note on (Any Two):a) Types of Errorsb) Convergent Methodsc) PDE

Group C

Attempt any TWO questions[2x10=20]

9. Write an algorithm and program to compute root of nonlinear equation using bisection method.

10. Given the data points:i012xi91625fi345Estimate the function value of f at $x = 14$ using cubic splines.

11. a) Solve the following system of equations using Gauss-Seidel method:2x - 7y - 10z = -175x + y + 3z = 14x + 10y + 9z = 7b) Write an expansion of Taylor’s Theorem. Solve the equation: $y'(x) = x^2 + y^2, \quad y(0) = 1$ for $x = 0.5$ using Taylor method.